

# Beyond the Given Wave Equation

From the RCP/LCP  $A_{z2,3}$  ODE (khaxn.pdf) to a validated theory of the non-Abelian Kelvin–Helmholtz instability, and its confrontation with GPU data

SU(2) Yang–Mills KH simulation program

2026-07-21

# Where we left off

## Already presented

khaxn.pdf: linearized Yang–Mills–Wong system in the  $a_x^a = 0$  gauge. Its eq. (23) closes the perturbation into a single decoupled 2nd-order wave equation for the circular combination  $a_z^+$  (i.e.  $A_{z2} + iA_{z3}$ ):

$$\left(-\partial_x^2 - \omega^2 + k_z^2 + \alpha^2(\phi_0^1)^2 - \alpha^2(A_{0z}^1)^2\right)a_z^+ = \frac{\alpha^2 V_{0z}}{\Omega + \alpha\phi_0^1} \left(V_{0z} - \frac{k_z + \alpha A_{0z}^1}{\omega + \alpha\phi_0^1}\right)a_z^+$$

**This talk covers everything derived and measured since.** Two more given results are used silently downstream — restated briefly next — then it's all new: the quartic → exact scalar reduction → growth ceiling → exact-action quantization → confinement-set wavelength selection → full GPU validation campaign → where Letter A actually stands today.

## The second given result (briefly): wkb.pdf

Expanding the step-potential form of the boxed equation to  $\mathcal{O}(v^2)$  gives a Schrödinger form  $\partial_x^2 a_z + Q_{\pm}(x) a_z = 0$  with a *parabolic* well,

$$Q_{\pm}(x) = Q_{0,\text{mean}} \pm \delta - |C_1|x^2 + C_2x^4, \quad |C_1| = \frac{\alpha^3 v^2}{2\omega^2},$$

and Bohr–Sommerfeld quantization of that harmonic core (the  $\pm\delta$  jump cancels between the two half-integrals) gives wkb.pdf eq. (33):

$$\boxed{Q_{0,\text{mean}} = (2n + 1)\sqrt{|C_1|}} \quad n = 0, 1, 2, \dots$$

### Physical picture

This is a harmonic-oscillator quantization: exact at  $k_z = 0$ , an uncontrolled extrapolation away from it. Everything from here derives *when* that extrapolation works, replaces it where it doesn't, and confronts both against real GPU data.

# Outline

- 1 From eq. (33) to the quartic,  $k_z=0$  anchor
- 2 The time-domain model
- 3 Six-field linearization
- 4 Local dispersion: tachyonic & drive branch
- 5 Exact scalar reduction: one ODE for  $a(x)$
- 6 The drive, resonance, growth ceiling
- 7 Quantization: models A and B
- 8 The dispersion curve: confinement-set peak
- 9 The quantization budget ( $\varepsilon_s$ )
- 10 Remaining theory, at pace
- 11 Data-quality: two bugs, found & fixed
- 12 Foundational validation anchors
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- 17 Where this leaves Letter A

## Clearing eq. (33) to a quartic

Substituting the harmonic-core definitions into eq. (33) and writing  $C \equiv (2n+1)\sqrt{\alpha^3/2} V_0$ :

$$\omega^2 - k_z^2 - \frac{\alpha^2 V_0 k_z}{\omega^2} = \frac{C}{\omega}$$

Multiply through by  $\omega^2$  ( $\omega = 0$  is not a root) and collect:

$$\omega^4 - k_z^2 \omega^2 - C \omega - \alpha^2 V_0 k_z = 0$$

the **WKB quartic**.  $\gamma = \max \text{Im } \omega$  over its four roots is the “ $\gamma_{\text{WKB}}$ ” quoted in every early program table.

### Caveat

An early implementation assumed  $\text{Re } \omega = 0$  (purely growing) and bisected on real  $\gamma$  — wrong away from  $k_z = 0$  by up to 50% at  $k_z = 0.25$ . The full complex quartic must be solved.

## The $k_z = 0$ limit: chromo-Weibel growth rate

Set  $k_z = 0$ :  $\omega^4 - C\omega = 0 \Rightarrow \omega(\omega^3 - C) = 0$ . The growing root is  $\omega = C^{1/3}e^{2\pi i/3}$ , giving

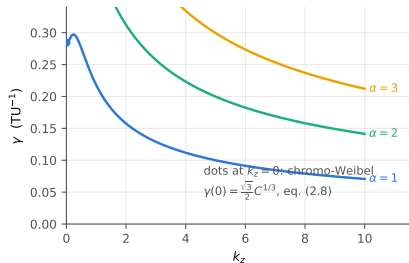
$$\gamma_{\text{Weibel}}(k_z=0) = \left( \sqrt{\frac{\alpha^3}{2}} V_0 \right)^{1/3} \sin 60^\circ$$

### Physical picture

**Not** a bunching instability — nothing varies along  $z$ . It's *filamentation*:  $A_{z1}$  rotates color-2/3 fields into each other while counter-streaming currents feed them back. Closing the loop  $E \rightarrow A \rightarrow Q \rightarrow J \rightarrow E$  gives gain  $\propto C/\gamma^3$ ; marginal self-amplification fixes  $\gamma \sim C^{1/3}$ , root geometry supplies  $\sin 60^\circ$ .

**Validated to 0.5%:**  $\alpha = 2$ ,  $V_0 = 0.1$ :  $\gamma_{\text{sim}} = 0.5039$  vs. formula 0.5065, over nine decades of growth — the strongest single anchor of the quartic, and the first number quoted to a referee (parameter-free).

# Quartic vs. reality: where it's valid, where it fails



- $k_z = 0$ : exact, validated to 0.5%.
- **Low**  $k_z$ : the quartic's drive  $\alpha^2 V_0 k_z / \gamma^2$  has no ceiling  $\Rightarrow$  overshoots the true branch by 10–20 $\times$ .
- **High**  $k_z$ : quartic balance gives  $\gamma^2 \approx \alpha^2 V_0 / k_z$ , a decline, while the true branch saturates near a ceiling  $\Rightarrow$   $\sim 25\%$  underestimate.

The diagnosis (why, precisely) has to wait for the exact reduction — coming up.

# The simulated system

For each color  $a \in \{1, 2, 3\}$ : fields  $E_x^a, E_z^a, B_y^a, A_z^a$ . For each beam  $s \in \{A, B\}$ : fluid  $n_s, p_{xs}, p_{zs}$ , color charge  $Q_s^a$ .

$$\begin{aligned}\partial_t n + \nabla \cdot (n \vec{v}) &= 0, & \partial_t p_x + \nabla \cdot (\vec{v} p_x) &= -n \sum_a Q^a (E_x^a + v_z B_y^a) \\ \partial_t p_z + \nabla \cdot (\vec{v} p_z) &= -n \sum_a Q^a (E_z^a - v_x B_y^a), & \partial_t Q^a &= \alpha v_z (\vec{Q} \times \vec{A}_z)^a \\ \partial_t E_x^a &= -\partial_z B_y^a - \alpha (\vec{A}_z \times \vec{B}_y)^a - J_x^a, & \partial_t E_z^a &= \partial_x B_y^a - J_z^a \\ \partial_t B_y^a &= \partial_x E_z^a - \partial_z E_x^a + \alpha (\vec{A}_z \times \vec{E}_x)^a, & \partial_t A_z^a &= -E_z^a\end{aligned}$$

Beams:  $v_{zA} = +V_0 \tanh \xi$ ,  $Q_A^1 = +1$ ;  $v_{zB} = -V_0 \tanh \xi$ ,  $Q_B^1 = -1$ ; frozen background  $A_{z1}(\xi) = -V_0 \ln \cosh \xi$ ,  $u(\xi) \equiv -\alpha A_{z1}(\xi) \geq 0$ .

# A gap worth closing: gauge counting

$SU(2)$  gauge freedom eliminates exactly *one* full spacetime component of  $A_\mu^a$  — not two. Dropping both  $A_x^a, A_y^a$  while also fixing  $A_0^a$  needs justification, not assumption.

one gauge choice  $\Rightarrow$  one component eliminated — not two.

- `khaxn.pdf` spends its choice on  $a_x^a = 0$ ;  $\phi$  survives, killed only by the Gauss *constraint*;  $a_y$  survives, decoupled from  $a_z$  by construction.
- This model spends a *different* choice,  $A_0^a \equiv 0$  (temporal gauge — exactly what  $\partial_t A_z = -E_z$  states).
- $A_y$  is then a *sector* choice (never populated);  $A_x$  is wired to *nothing* in any equation of motion — a complete spectator either way.

## Physical picture

One knob, one turn: two different single choices, not one spent twice.  $E_x$  carries exactly  $\phi$ 's slaved role ( $\gamma e_x = -i\Omega_A b$ , purely algebraic).

# Precession source from Wong's equation

Wong's equation, non-relativistic limit, temporal gauge, only  $A_z^b \neq 0$ :

$$\frac{dQ^a}{dt} = -g \epsilon^{abc} v_z A_z^b Q^c = +\alpha v_z (\vec{Q} \times \vec{A}_z)^a$$

exactly the model's source;  $|\vec{Q}|$  conserved  $\Rightarrow Q_{A,B}^1 = \pm 1$  stays a unit adjoint vector. **Energy budget** (derivation skipped): non-Abelian cross terms do *not* cancel —

$$\left. \frac{dE}{dt} \right|_{\text{NA}} = -2\alpha \int \vec{A}_z \cdot (\vec{B}_y \times \vec{E}_x)$$

## Physical picture

A frozen field is an infinite battery. In the tachyonic region this term *feeds* the wave without limit (Nielsen–Olesen-type decay). In the shear well it's a drain; the shear branch draws on the beams' kinetic energy instead — a finite, self-consistent reservoir.

## Circular pairs: color rotation is a phase

Combine color-2/3 perturbations:  $b = B_y^2 + iB_y^3$ ,  $e_x = E_x^2 + iE_x^3$ ,  $e_z = E_z^2 + iE_z^3$ ,  $a = A_z^2 + iA_z^3$ ,  $q_{A,B} = Q_{A,B}^2 + iQ_{A,B}^3$ , ansatz  $\propto e^{ik_z z + \gamma t}$ . For background  $\vec{A}_z = (A_{z1}, 0, 0)$ , any commutator  $(\vec{A}_z \times \vec{X})^2 + i(\vec{A}_z \times \vec{X})^3 = iA_{z1}X^c$ : a cross product with the background acts on a circular pair as multiplication by  $iA_{z1}$ .

$$\Omega_A(\xi) = k_z + \alpha A_{z1} = k_z - u(\xi), \quad \Omega_F(\xi) = k_z - \alpha A_{z1} = k_z + u(\xi)$$

### Physical picture

The color-2/3 waves are *charged* (charge  $\pm 1$ ) under the residual  $U(1)$  about the color-1 axis;  $\alpha A_{z1}$  shifts their momentum like an ordinary gauge potential,  $k_z \rightarrow k_z \pm \alpha A_{z1}$ . Ampère sees  $\Omega_A$ , Faraday sees  $\Omega_F$  — the split is the fingerprint of the fixed  $A_x = 0$  constraint.

# The six-field eigenvalue system

$$\begin{array}{ll} \gamma \mathbf{b} = -i\Omega_F \mathbf{e}_x + \mathbf{e}'_z & \gamma \mathbf{a} = -\mathbf{e}_z \\ \gamma \mathbf{e}_x = -i\Omega_A \mathbf{b} & \gamma \mathbf{q}_A = i\alpha v \mathbf{a} - i v \Omega_A \mathbf{q}_A \\ \gamma \mathbf{e}_z = \mathbf{b}' + v(\mathbf{q}_A - \mathbf{q}_B) & \gamma \mathbf{q}_B = i\alpha v \mathbf{a} + i v \Omega_A \mathbf{q}_B \end{array}$$

This is exactly what `ym_eigenmode.py` solves. Validated against the 2D simulation at 1–4% on the  $V_0 = 0.05$  validation stack ( $k_z \leq 5$ ); 4–9% median across the full integer- $k_z$  map; 6–10% archive-wide clean core. **Correspondence check:** eliminating  $\mathbf{e}_x, \mathbf{e}_z, \mathbf{q}_A, \mathbf{q}_B, \mathbf{b}$  reproduces exactly the beam-response denominators  $\Omega \pm \alpha \Phi_0^1$  of the boxed equation on slide 2, under  $\omega = i\gamma$  — the time-domain model linearizes into *precisely* the given frequency-domain theory, modulo two stated scope truncations (no background  $\phi_0^1$ ; no  $a_y^\pm$  sector). **Symmetry:** helicity conjugation ( $A_{z1} \rightarrow -A_{z1}$ ) combined with  $\xi \rightarrow -\xi$  maps the system onto itself — every eigenmode on one side of the layer has a mirror partner of the opposite polarization on the other side, same  $\gamma$ .

## Block factorization at $k_x = 0$

Freezing the background locally, the system splits into two independent blocks:

$$\underbrace{\gamma b = -i\Omega_F e_x, \quad \gamma e_x = -i\Omega_A b}_{\text{EM block}}$$

$$\underbrace{\gamma e_z = v(q_A - q_B), \quad \gamma a = -e_z, \quad \gamma q_{A,B} = i\alpha v a \mp i v \Omega_A q_{A,B}}_{\text{charge block}}$$

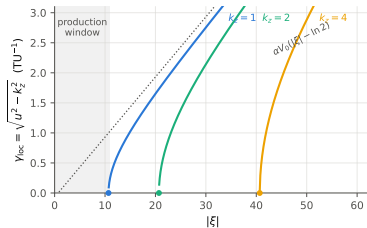
# EM block: the tachyonic (Nielsen–Olesen) branch

Multiplying the EM block:

$$\gamma^2 = -\Omega_A \Omega_F = \alpha^2 A^2 - k_z^2$$

Negative wherever the background shift exceeds the wavenumber  $\Rightarrow \gamma^2 > 0$ : the charged-vector-on-strong-background instability of the Nielsen–Olesen class.

$$\gamma_{\text{loc}}(\xi) \simeq \sqrt{(\alpha V_0)^2 (|\xi| - \ln 2)^2 - k_z^2} \quad \text{real for} \quad |\xi| > \xi_{\text{crit}} \simeq \frac{k_z}{\alpha V_0} + \ln 2$$



## Consequence: high- $\alpha$ window collapse

A window of radius  $\xi_w$  excludes the tachyon only if  $\xi_w < \xi_{\text{crit}}$ . As  $\alpha$  grows at fixed  $k_z, V_0$ :  $\xi_{\text{crit}} \rightarrow \ln 2$  — eventually *no* window both contains the shear well and excludes the tachyon.

### Caveat

This is the mechanism of the early tachyonic blow-ups at  $\alpha \geq 6$ : at  $\alpha = 6, V_0 = 0.05, k_z = 2$ :  $\xi_{\text{crit}} = 7.4$ , already inside the vetted windows. The failure maps to  $\alpha$ , exactly as predicted — this is now known to be only *part* of the story (resolution addendum, later).

Same mechanism, evaluated at layer scale ( $|A_{z1}| \sim V_0$ ) rather than the outer plateau, gives the “precession cascade”  $\gamma_{\text{casc}} \approx \alpha V_0$  that motivated full eigenmode seeding (later in this talk).

## Charge block: the local drive branch

Solving the two precession equations and closing through  $e_z, a$ :

$$q_A - q_B = \frac{2\alpha v^2 \Omega_A}{\gamma^2 + v^2 \Omega_A^2} a \quad \Rightarrow \quad \boxed{\gamma^2 = -g, \quad g(\xi; \gamma) = \frac{2\alpha v^3 \Omega_A}{\gamma^2 + v^2 \Omega_A^2}}$$

### Physical picture

Two driven pendulums. The potential  $a$  shakes each beam's color charge; each precesses at Doppler-shifted  $\pm v \Omega_A$ . Only their *difference* radiates back into  $a$  through the current — growth is a phase-locking resonance between the wave and the beams' color precession. This is the local ( $k_x \rightarrow 0$ ) limit of the drive that reappears, unmodified, in the exact reduced ODE next.

# The reduction — algebra all the way down

The six-field system has only *one*  $x$ -derivative chain ( $e'_z \rightarrow b \rightarrow b'$ ); everything else is algebraic. Eliminating  $e_x, e_z, q_A, q_B, b$ :

$$b = -\frac{\gamma^2 a'}{\mathcal{D}}, \quad \mathcal{D}(x) \equiv \gamma^2 + \Omega_A \Omega_F = \gamma^2 + k_z^2 - u(x)^2$$

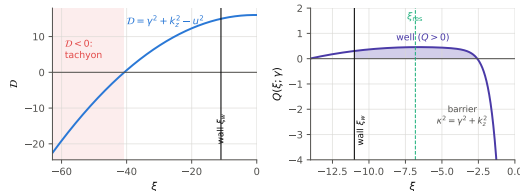
Assembling Ampère- $z$  gives *exactly one* second-order ODE — the new theory's counterpart of the boxed equation on slide 2, now derived independently and exactly from the six-field system:

$$\left( \frac{\gamma^2 a'}{\mathcal{D}} \right)' = (\gamma^2 + g) a, \quad g(x) = \frac{2\alpha v^3 \Omega_A}{\gamma^2 + v^2 \Omega_A^2}$$

## Physical picture

Three powers of  $v$  in  $g$ : the wave must (i) shake the charges, (ii) make the two beams respond *differently* (opposite Doppler rotation), (iii) collect the response as a current. The drive dies as  $v^3$  toward the layer center — the shear branch is anchored to the flow, not the field maximum.

# Anatomy of the reduced equation



- $D$ : sign change separates trapped shear ( $D > 0$ ) from tachyonic outer ( $D < 0$ ) — *one* mechanism, two fates.
- $g$ : the only drive of the shear branch, the local-drive function.
- WKB form  $a \propto \exp(\pm i \int \sqrt{Q} dx)$ , exact local wavenumber  $Q = -D(\gamma^2 + g)/\gamma^2$ .

**Exactness verified:** discrete Newton iteration reproduces six-field ARPACK eigenvalues to rel. diff.  $< 10^{-10}$ , 7 points,  $\alpha = 1-3$ .

## The saturated drive $G(w)$ and its resonance

In the saturated region ( $v = \pm V_0$ ,  $u = \alpha V_0(|\xi| - \ln 2)$ ), on the driven side:

$$-g = G(w) = \frac{2\alpha V_0^3 w}{\gamma^2 + V_0^2 w^2}, \quad w \equiv \Omega_A = k_z - u(\xi)$$

Maximizing over  $w$ :  $\frac{dG}{dw} = 0 \Rightarrow w^* = \gamma/V_0$  — the mode grows fastest where the *beam-frame precession frequency* matches the growth rate itself. Peak value:

$$G_{\max} = G(w^*) = \frac{\alpha V_0^2}{\gamma}$$

## The growth ceiling: $\gamma^3 \leq \alpha V_0^2$

A confining well needs  $G > \gamma^2$  somewhere; since  $G \leq G_{\max} = \alpha V_0^2 / \gamma$ :

$$\gamma^3 \leq \alpha V_0^2$$

independent of  $k_z$ ,  $\varepsilon_s$ , and window geometry. **Two independent derivations agree** (well-existence, and the local-drive branch  $\gamma^2 = G(w^*)$  solved directly).

### Physical picture

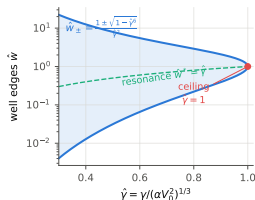
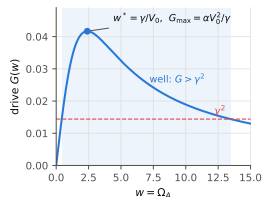
A self-limiting resonance. The beam response is a Lorentzian in local precession frequency, width set by  $\gamma$  itself: the faster the mode grows, the broader the band it averages over, the weaker the peak drive it can extract. The ceiling is the fixed point — the mode grows exactly as fast as the drive it leaves itself.

**Measured:** dominant-branch peaks reach 95–99% of  $(\alpha V_0^2)^{1/3}$  in every validated series (empirical confirmation later, T2.5).

# Natural units: a universal well

$\Gamma \equiv (\alpha V_0^2)^{1/3}$  (rate scale),  $K \equiv (\alpha/V_0)^{1/3}$  (wavenumber scale),  $\gamma = \Gamma \hat{\gamma}$ . The drive becomes *universal* — no  $\alpha$ ,  $V_0$ ,  $\varepsilon_s$  remain:

$$G(w) = \Gamma^2 \hat{G}(\hat{w}), \quad \hat{G}(\hat{w}) = \frac{2\hat{w}}{\hat{\gamma}^2 + \hat{w}^2}$$



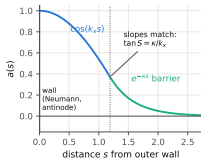
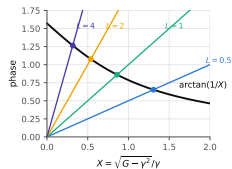
$\hat{\gamma} \leq 1$  — a *third* derivation of the ceiling. **Rates scale as  $(\alpha V_0^2)^{1/3}$ , wavenumbers as  $(\alpha/V_0)^{1/3}$  — no function of  $\alpha V_0$  alone works.**

Well edges  $\hat{w}_\pm = (1 \pm \sqrt{1 - \hat{\gamma}^6})/\hat{\gamma}^2$ , real only for

# A quarter-wave well

The well is shallow: measured action  $S = 0.31\pi - 0.46\pi$  — less than a quarter wavelength. Naive Bohr–Sommerfeld fails; the connection condition at the sharp inner “knee” does the work instead:

$$\tan S = \frac{\kappa}{k_{\text{edge}}}, \quad \kappa = \sqrt{\gamma^2 + k_z^2}$$



## Physical picture

A surface-mode-like state, not a cavity mode: rays hug the ceiling (a small drop in  $\gamma$  buys all the depth needed), while the *peak position* is set entirely by window geometry.

## Model A (closed form) and Model B (exact action)

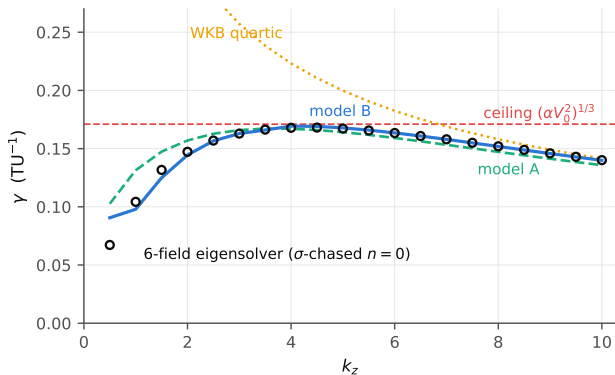
**Model A:** square-well approximation ( $u \ll k_z$ ,  $\mathcal{D} \approx \gamma^2 + k_z^2$  const.) closes on a single transcendental equation  $LX = \arctan(1/X)$  ( $L = \varepsilon_s(\xi_w - \xi_0)k_z$ ), giving

$$\gamma^2 = \frac{1}{2} \left[ -V_0^2 k_z^2 + \sqrt{V_0^4 k_z^4 + \frac{8\alpha V_0^3 k_z}{1+X^2}} \right]$$

Validated to  $\sim 3\%$  near/above the peak; locates the peak to  $\pm 0.5$  in  $k_z$ . **Model B:** keeps the *exact*  $Q(x; \gamma)$ , root-finds  $S(\gamma) = \int \sqrt{Q} dx = \Phi(\gamma)$  (connection phase, including Airy  $\pi/4$  terms at soft turning points).

**Matches  $\sigma$ -chased  $n=0$  eigenvalues to  $\leq 2\%$  (median  $\leq 0.2\%$ ) across 12 series,  $k_z \geq 1.5$ .**

# The full theory chain, one dispersion curve



Open circles:  $\sigma$ -chased 6-field eigenvalues. Solid: Model B. Dashed: Model A. Dotted: the old WKB quartic — overshooting exactly as diagnosed earlier.

# A methodological trap: the overtone ladder

$S(\gamma)$  decreasing in  $\gamma \Rightarrow$  a *ladder* of well overtones. A shift-invert eigensolver returns whatever eigenvalue is nearest its target  $\sigma$  — the legacy default lands mid-ladder wherever the old quartic underestimates.

## Caveat

**Discovered via this theory, then falsified experimentally:** at  $(\alpha, V_0, k_z) = (1, 0.05, 4)$ , sponge 20 — the *same* physical point, seeded two different ways:

| seed                                          | eigenvalue | measured plateau |
|-----------------------------------------------|------------|------------------|
| default- $\sigma$ (an $n \approx 3$ overtone) | 0.0818     | <b>0.0808</b>    |
| $\sigma = 0.14$ (true $n = 0$ )               | 0.1309     | <b>0.1287</b>    |

The simulation faithfully grows whichever mode it's handed. “Sim matches eigensolver” certifies numerics, *not* dominant-mode selection. All reference data from 2026-07-19 onward use  $\sigma$ -chasing.

## No intrinsic $k_z$ maximum

In an unbounded domain, nothing stops the mode following the resonance surface outward as  $k_z$  grows:  $G_{\max} = \alpha V_0^2 / \gamma$  is independent of  $k_z$ , so  $\gamma(k_z)$  rises monotonically to the ceiling while the mode migrates outward.

**Directly demonstrated:**  $\gamma$  climbs to  $0.9955 \Gamma$  by  $k_z = 10$  with *no turnover*, mode peak at  $\xi = -36$  (loose cut 40).

### Physical picture

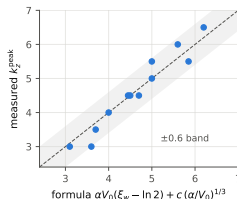
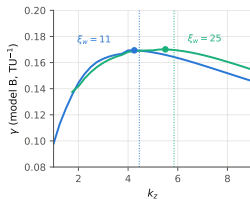
The measured non-monotonic dispersion curve is therefore **windowed physics**, not an intrinsic wavelength selection. This is the single most counter-intuitive statement of the whole theory.

# The peak: resonance surface meets the window

Resonance surface  $\xi_{\text{res}} \simeq (k_z - \gamma/V_0)/(\alpha V_0) + \ln 2$  moves outward *linearly* with  $k_z$ . Setting  $\xi_{\text{res}}(k_z) = \xi_w$ ,  $\gamma \approx \Gamma$  at the crossover:

$$k_z^{\text{peak}} \simeq \alpha V_0(\xi_w - \ln 2) + c (\alpha/V_0)^{1/3}, \quad c = 1.0 \pm 0.1$$

**Both terms  $\varepsilon_s$ -independent:** coupling and containment radius select the wavelength, together — not



the shear width.

## Roll-off, and why the data used to say “ $k_z^{\text{peak}} \approx 2\alpha$ ”

Beyond the peak (window-truncated well, high- $k_z$  limit of Model A):

$$\boxed{\gamma \simeq \sqrt{2\alpha V_0/k_z}} \quad (\gamma \propto k_z^{-1/2} \text{ decline, matches data})$$

### Physical picture

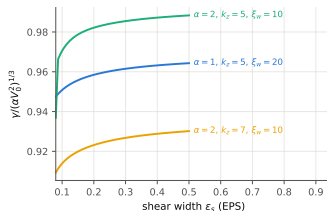
Over the historical survey grid, window radii shrank as  $\alpha$  grew ( $\alpha V_0 \xi_w \approx 1$ ) — the sub-linear resonance term plus a shrinking window term *mimics* a linear trend across a factor-5 range of  $\alpha$ . “ $k_z^{\text{peak}} \approx 2\alpha$ ” is a serviceable *survey-box mnemonic*; the boxed formula above is the actual law.

# Nondimensionalizing the action

Neither the ceiling nor the peak formula contains  $\varepsilon_s$  — where does it enter? Changing variables  $x = \varepsilon_s \xi$ ,  $w = k_z - u$ , scaling out the drive:

$$S = \frac{\varepsilon_s}{V_0 (\alpha V_0^2)^{1/3}} \hat{I}(\hat{k}, \hat{\gamma}), \quad \hat{I} = \int_{\hat{w}\text{-well}} \sqrt{\hat{Q}} d\hat{w}$$

$\hat{I}$  is *universal* —  $\varepsilon_s$ -free. The shear width enters the eigenvalue problem **only** as a multiplicative budget



on the action.

## The $P$ -group: $\varepsilon_s$ buys action

Larger  $\varepsilon_s$  (fixed window, physics) enlarges  $S$  at given  $\gamma$ ; since  $S$  decreases in  $\gamma$ , quantization pushes  $\gamma$  *up* toward the ceiling — the sign of the measured 10–39% rise of  $\gamma$  from  $\varepsilon_s = 0.10 \rightarrow 0.45$ .

$$P \equiv \frac{(\alpha V_0^2)^{1/3}}{\varepsilon_s} \quad (\text{the “quantization budget”})$$

### Physical picture

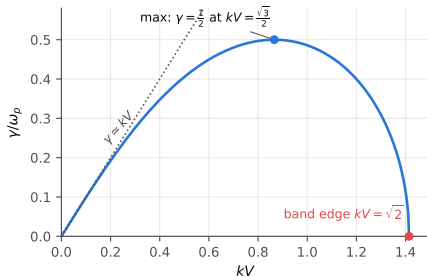
$\varepsilon_s$  buys action, not a wavelength shift. Stretching the layer stretches the well in physical  $x$  while its depth/shape in  $\hat{w}$  is unchanged — so at fixed  $\gamma$  the mode accumulates more phase, and the connection condition lets  $\gamma$  float up. This is why  $\varepsilon_s$ 's effect is a fractional *rise* of  $\gamma$  everywhere, not a shift of  $\Gamma, K$  — and it's the theory this talk's GPU section will confront directly (T1.1).

# Cold two-stream rates (fixes the filter design)

Away from the layer, the beams support an electrostatic color-1 two-stream mode:

$$1 = \frac{1}{(\omega - kV_0)^2} + \frac{1}{(\omega + kV_0)^2} \Rightarrow \boxed{kV_0 < \sqrt{2}} \text{ (unstable band)}$$

$$\gamma^2 = \sqrt{4(kV_0)^2 + 1} - (kV_0)^2 - 1$$



Max  $\gamma = \frac{1}{2}$  at  $kV_0 = \sqrt{3}/2$ ; band edge  $kV_0 = \sqrt{2}$ . At  $V_0 = 0.1$ :  $k < 14.1$  — **exactly** the documented  $k_{ts} \approx 14$  band edge that fixed the production bandpass filter, and the spatial profile that ruled the two-stream *out* as the late-catastrophe driver.

# Nonlinear theory I: the density cavitation threshold

Why  $V_0 \gtrsim 0.08$  hits a hard wall: a growing compressible mode carries  $\delta n \propto p/\gamma$ ; cavitation ( $n \rightarrow$  code floor) occurs at

$$p_{\text{cav}} \sim \gamma / \max(k_z, k_x^{\text{eff}})$$

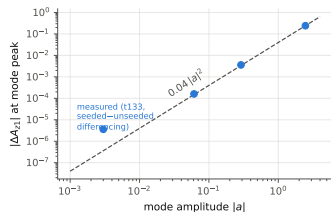
## Physical picture

An *amplitude* cliff, not a time cliff. A growing wave in a compressible fluid carries a density modulation  $\propto$  momentum/ $\gamma$ ; the quadratic (ponderomotive) beat then pushes plasma out of high-field regions at  $2\gamma$ . This explains why the failure zone maps to  $V_0$  (drive strength) and cares little about window design or mechanism — and why the only safe practice at high  $V_0$  is to stop once the linear plateau is banked.

# Nonlinear theory II: the back-reaction on $A_{z1}$

Any single-helicity beat is  $z$ -independent ( $k_z = 0$ ), growing at  $2\gamma \Rightarrow$  quadratic law:

$$\Delta A_{z1} \simeq -0.04 |a|^2 \quad (\text{measured over 4 decades, endpoint exact})$$

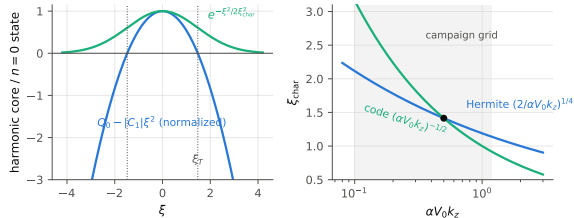


## Physical meaning

Sign *deepens* the background — no depletion. Waves weigh on their background like radiation pressure; the frozen- $A_{z1}$  approximation is vindicated, not just assumed — the last escape route for that caveat closes.

# Warm closure and eigenmode seeding

**Warm isothermal closure:**  $\delta F_i = -T \partial_i n$  ( $T = 0 \Rightarrow$  cold model exactly). Stabilizes the two-stream channel when  $v_{th} \gtrsim V_0$  — but *cannot* stabilize  $k_z = 0$  chromo-Weibel (anisotropy-driven, not pressure-limited; confirmed: final  $k_z=0$  amplitude shifts only  $\sim 11\%$  over  $v_{th}/V_0 = 0-3$ ). **Eigenmode seeding** (why single-field seeding fails): the harmonic core's  $n = 0$  Hermite ground state gives the natural seed width  $\xi_{char} = (2\gamma^2/\alpha^3 V_0^2)^{1/4}$ . An  $A_z$ -only seed can overlap a *different*, co-located, slower mode (measured:  $\gamma = 0.060$  vs. target 0.122) — or lose a race against the layer-scale cascade  $\gamma_{casc} \approx \alpha V_0$ . Fix: seed *all six fields* with the solver's dominant eigenvector.



# Theory summary: the validation ledger

| Derived result                       | Numerical anchor                                               |
|--------------------------------------|----------------------------------------------------------------|
| $k_z=0$ chromo-Weibel rate           | 0.5%                                                           |
| Six-field system                     | 1–4% (validation stack); 4–9% full-map median                  |
| Exact scalar reduction               | rel. diff. $< 10^{-10}$ , 7 points                             |
| Tachyonic law                        | analytic = eigensolver = CUDA at all outer peaks               |
| Ceiling $\gamma^3 \leq \alpha V_0^2$ | peaks at 95–99% in every series; T2.5: $0.977 \pm 0.011$       |
| Model B quantization                 | $\leq 2\%$ (median $\leq 0.2\%$ ), 12 series                   |
| Peak formula                         | $\pm 0.6$ in $k_z$ , 11 series; window test confirmed directly |
| Two-stream band                      | $k_{ts} \approx 14$ at $V_0 = 0.1$ (documented)                |
| Back-reaction law                    | endpoint $-0.242$ vs. $-0.242$                                 |
| Warm closure                         | warm vs. cold theory: 1%                                       |

**That's the theory. Now: does it survive contact with the GPU?**

# Bug 1 — the pipeline measured the *conjugate* helicity

2026-07-04 stdlib-DFT rewrite: a sign error computed the projection onto  $e^{-ik_z z}$  instead of  $e^{+ik_z z}$ .

## Caveat

For a pure single-helicity seed the conjugate projection is *identically zero* — the **empty channel**, not a noisier right answer. **Blast radius**: of 7,909 archived timeseries, **96% (7,589)** start below the float32 noise floor. Raw field dumps are deleted after processing: **these runs cannot be re-extracted**, only rerun.

## Physical picture

Why unnoticed for two weeks: rounding slaves a small image of the *true* mode into the empty channel, inheriting  $\gamma$  almost exactly — looks like a clean exponential,  $R^2 = 1.000$ . The two only visibly diverge where a second fast channel competes for dominance — precisely prior audits' worst outliers.

Fixed in `remote_timeseries.py` (now emits the conjugate amplitude as an explicit diagnostic).

## Bug 2, and a related numerical rule

**Bug 2:** `gen_suspectfix_campaign.py` defined half/fine-tier box parameters but never emitted them into the `.ini` — every half/fine-tier “suspectfix” run silently ran on the *default*  $L_z = 2\pi$  box, at  $k_z = \text{mode-index}$  (not  $/2$  or  $/8$  as labeled).

### Caveat

**The two bugs compounded:** Bug 1 returned the wrong channel, Bug 2 relabeled a  $k_z = 1$  measurement as  $k_z = 0.5$ . A whole population of “anomalies” attributed to genuine sub- $k_z = 1$  physics (a floor, a 68% error block, a two-branch structure) was bookkeeping on top of a measurement of nothing.

**Rule found during the fix:** keep  $k_{\text{suppress,hi}} \leq N_z/4.5$  — a band-edge scan found the kept mode’s rate collapsing to  $\sim 0.001$  as the DFT bandpass approaches the grid’s Nyquist-adjacent dispersive tail. **Bottom line (post-fix):** sim and eigensolver agree to  $\leq 8\%$  at every anchor tested, every box tier, integer and fractional  $k_z$ .

## The $k_z = 0$ anchor, again — now cross-checked

$\alpha = 2.0, V_0 = 0.1$ :  $\gamma_{\text{sim}} = 0.5039$  vs. formula 0.5065 — 0.5%. **Independent cross-check**: the unfrozen self-consistency reference point ( $\alpha = 1, V_0 = 0.05$ ) reproduces the *same* formula ( $\gamma = 0.284$ , exact match) on a completely different (unfrozen, color-1-live) configuration, three weeks and many campaigns later.

### Physical picture

Why lead a referee here: exercises every non-Abelian term at once, at the one point where the harmonic-core approximation is essentially exact — no theory-side slack to hide a code bug behind.

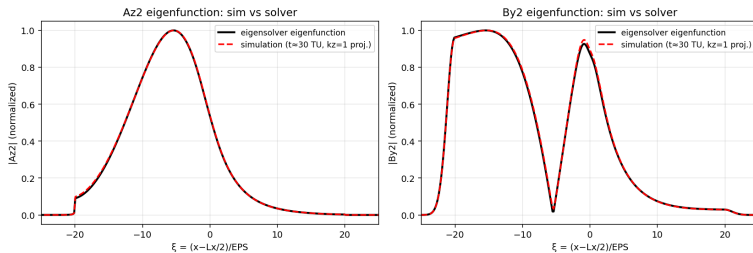
# The referee-proofing battery

| Experiment                       |   | Headline                                                      |
|----------------------------------|---|---------------------------------------------------------------|
| T2.4 linearity                   | ✓ | $\gamma$ invariant to seed amplitude, $100\times$ range       |
| T2.2 eigenfunction overlay       | ✓ | sim vs. solver correlation = 1.0000                           |
| T2.3 Gauss law                   | ✓ | residual $\sim 10^{-5}$ , decreasing, localized               |
| T2.1 complex frequency           | ✓ | dominant branch purely growing ( $\lesssim 6\%$ )             |
| T2.5 dimensionless collapse      | ✓ | $\gamma_{\text{peak}}/(\alpha V_0^2)^{1/3} = 0.977 \pm 0.011$ |
| T2.7 sponge $\rightarrow \infty$ | ✓ | saturates $\xi_w \approx 16$ ; $\xi_w = 6$ : $-28.5\%$        |
| T2.8 resolution extremes         | ✓ | $< 1\%$ low corner, <b>revised below</b>                      |

All eight run on free teaching nodes at reference  $\alpha = 1$ ,  $V_0 = 0.05$ ,  $k_z = 1$  (2026-07-19); the eighth — the overtone-reseed falsification — is the ladder test shown earlier.

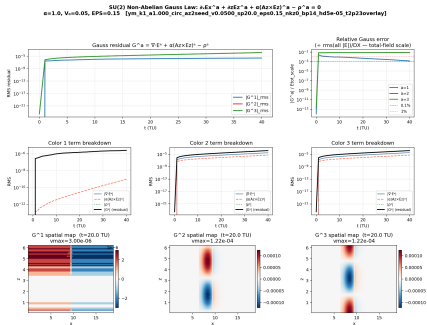
## T2.2: the “money figure”

T2.2 Eigenfunction overlay —  $\alpha=1$ ,  $V_0=0.05$ ,  $k_z=1$  ( $A_z$  profile corr=1.0000,  $\xi_{\text{peak}}$ : sim=-5.4 vs solver=-5.4)



$A_z^2$  profile correlation  
= 1.0000.  $B_y^2$  — seeded at only  $0.008\times$  the  $A_z$  amplitude, effectively grown from zero via the activation chain — reproduces the solver's non-trivial **double-lobe shape**, node at  $\xi \approx -5$ : an emergent structural match, not just a matched growth rate.

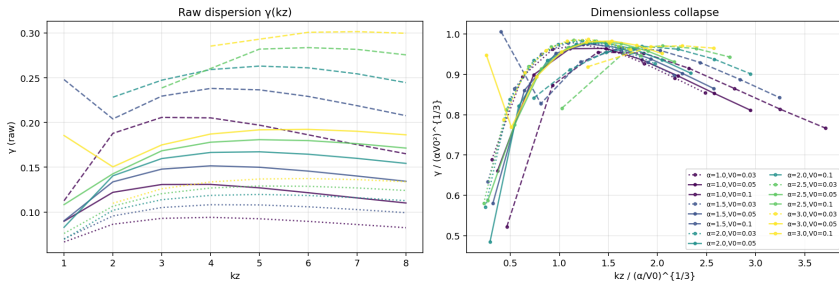
## T2.3: Gauss-law conservation



Leapfrog was never designed to preserve the covariant constraint exactly (same as Yee-grid FDTD, Abelian case). Absolute residual  $\sim 10^{-5}$ ; color-1 *relative* residual  $\sim 10^{-3}$  and **decreasing**; divergence term dominates, non-Abelian term subdominant, source negligible.

## T2.5: the dimensionless collapse

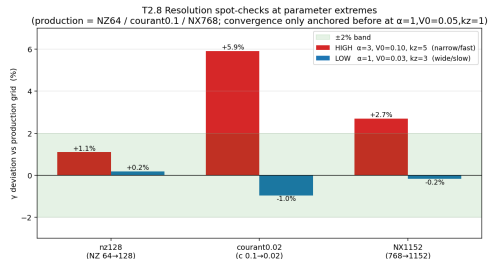
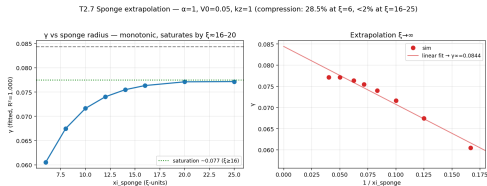
T2.5 Dimensionless collapse of the non-Abelian KH dispersion ( $\sigma$ -chased eigensolver)



$$\gamma_{\text{peak}}/(\alpha V_0^2)^{1/3} = 0.977 \pm 0.011 \quad (1.1\% \text{ spread, } 10 \times \text{ range in } \alpha V_0)$$

The single most quantitative confirmation in the program: 15 dispersion curves fold onto one master curve — direct proof  $\alpha$ ,  $V_0$  enter through  $(\alpha V_0^2)^{1/3}$  and  $(\alpha/V_0)^{1/3}$ , not the naive product  $\alpha V_0$ .

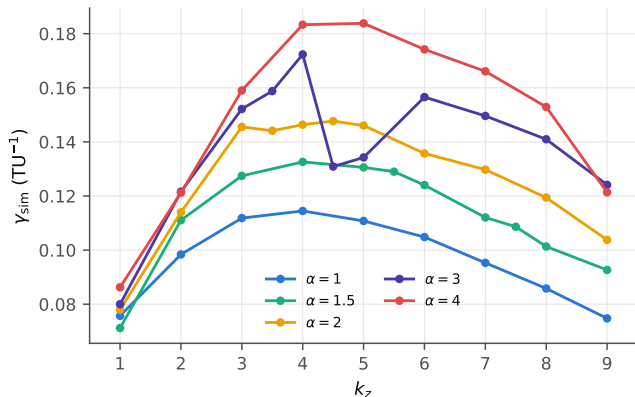
# T2.7 and T2.8, with a correction



## Caveat

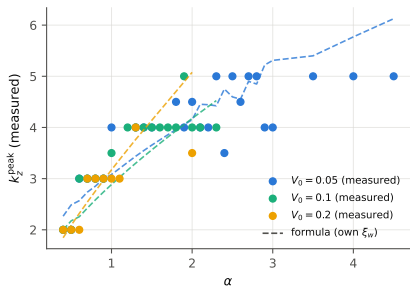
**T2.8 revised (2026-07-21, resolution/Lx campaign):** the original “3–6% at the high- $\alpha V_0$  corner” tested each grid axis (NZ, courant, NX) *separately*. Testing them *together* (NX=1152 + courant=0.05): **+6.6%**; all three axes together: **+11.6%** — larger than the naive additive estimate (+9.7%). **Revise the quoted figure to ~10–12% at the high- $\alpha V_0$  end.** Corrections compound, they don’t saturate.

# Non-monotonic dispersion, real data



$\gamma_{\text{sim}}(k_z)$  at  $V_0 = 0.05$ , five values of  $\alpha$  — from the trimmed integer- $k_z$  map (1087/1134 points), **not** a theory curve. Every curve peaks at intermediate  $k_z$ ; peak location climbs from  $k_z^{\text{peak}} \approx 2$  ( $\alpha = 1$ ) to  $\approx 4-5$  ( $\alpha = 3-4$ ), at fixed  $\varepsilon_s = 0.15$ . Classical KH also has a peaked  $\gamma(k_z)$  — but its peak is fixed by the shear width ( $k_z^{\text{peak}} \sim 0.4/\varepsilon_s$ ). A peak that moves with the *coupling* at fixed flow profile has no classical analogue.

$k_z^{\text{peak}}$  tracks the confinement formula, not “ $2\alpha$ ”



| $(\alpha, V_0, \xi_w)$ | meas. | formula |
|------------------------|-------|---------|
| $(1, 0.05, 20)$        | 3.5   | 3.7     |
| $(2, 0.05, 11)$        | 4.5   | 4.5     |
| $(3, 0.03, 5)$         | 5.5   | 5.0     |
| $(5, 0.03, 5)$         | 6.5   | 6.2     |
| $(1, 0.20, 10)$        | 3.0   | 3.6     |

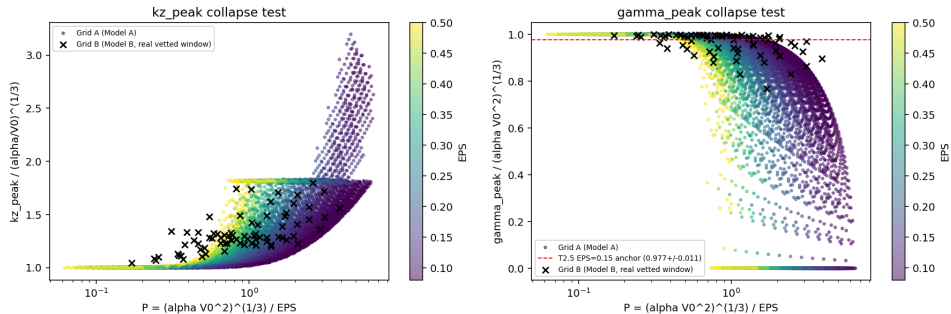
**Window test** ( $\alpha = 2, V_0 = 0.05$ ): cut 11  $\rightarrow$  25 moves the peak 4.5  $\rightarrow$  5.5 (formula: 4.45  $\rightarrow$  5.85).

## Physical picture

Two window radii, same  $(\alpha, V_0)$ , two different (both correct)  $k_z^{\text{peak}}$  — **coupling and confinement radius jointly select the wavelength; the flow profile does not.**

# EPS-independence: the theory-first check

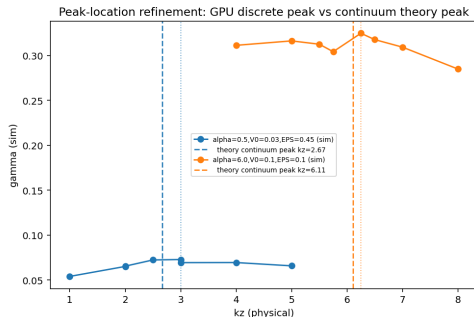
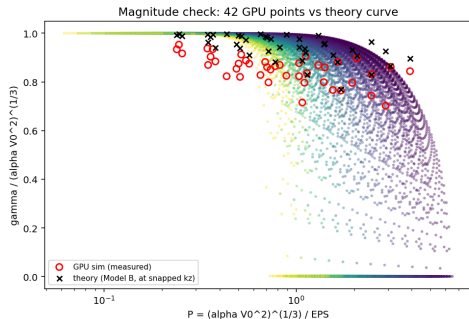
EPS-scan v2 Phase 0: does EPS enter only through  $P=(\alpha V_0^2)^{1/3}/\text{EPS}$ ?



CPU-only test of the  $P = (\alpha V_0^2)^{1/3}/\varepsilon_s$  collapse hypothesis *before any GPU time spent* —  $\alpha$  up to 10,  $V_0 = 0.01$ – $0.20$ . Both  $k_z^{\text{peak}}/K$  and  $\gamma_{\text{peak}}/\Gamma$  are close to single-valued functions of  $P$  alone. Directly explains the two open T1.1 puzzles as one curve viewed at different  $P$ .

# EPS-independence: GPU confirmation (91 runs)

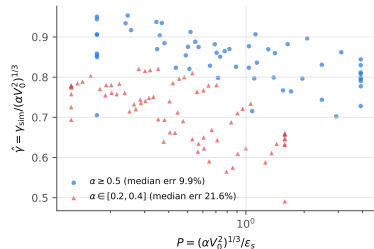
EPS-scan v2 Phase 2: GPU validation of the Phase 0 theory collapse



$\alpha = 0.5\text{--}6$ ,  $V_0 = 0.03\text{--}0.10$ ,  $\varepsilon_s = 0.10\text{--}0.45$ : median relative error **9.9%** (90th pct. 17.0%) vs. the  $P$ -collapse theory. Two 9-point peak-location cascades land within 0.14–0.33 in  $k_z$  of the continuum theory peak.

**This closes roadmap item T1.1.**

# Extension to low $\alpha$ : the collapse degrades, doesn't break



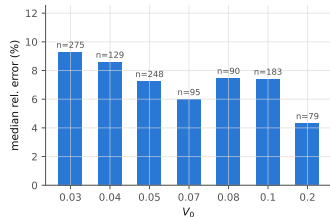
75 more runs,  $\alpha \in \{0.2, 0.25, 0.3, 0.4\}$  ( $\alpha \leq 0.15$  excluded — known noise floor). Median relative error **21.6%** (double the  $\alpha \geq 0.5$  figure), with real  $\alpha$ -scatter at matched  $P$  not seen at higher  $\alpha$ .

## Physical picture

$P$ -collapse is right down to  $\alpha = 0.2$ , with a sub-leading  $\alpha$ -dependence visible near the noise floor — not a theory failure, a measurement of where the single-group approximation needs a correction.

# Full-grid accuracy: 1087/1134 points (95.9%)

Grid:  $k_z = 1..9$ ;  $\alpha = 0.3\text{--}6.0$  (18 values);  $V_0 \in \{0.03, \dots, 0.20\}$  (7 values) — 1134 points.



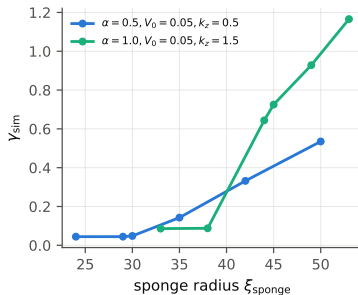
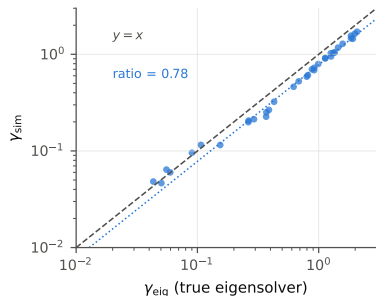
Median relative error vs.  $\sigma$ -chased eigensolver, plateau-confirmed core:

**4.3–9.4% depending on  $V_0$** , broadly improving with  $V_0$  (best at  $V_0 = 0.2$ ).

## Caveat

**New failure corner:** all 47 unmeasured points sit at  $V_0 = 0.2, \alpha \geq 3.5$  — a fast halt, mechanism not yet determined. Report as *unmeasured*, not measured-zero.

# The outer tachyonic branch: 46-point confirmation



50 runs ( $\alpha \leq 2.0$ ), 46/50 fit cleanly, benchmarked against the *true* eigensolver rate (not the sponge-edge estimate,  $5\text{--}10\times$  too high): **median error 22.2%, ratio 0.78** — the same  $\sim 20\text{--}28\%$  sponge-compression signature seen elsewhere, not scatter. Two sponge-depth cascades confirm the law directly:  $12\text{--}13.5\times$  rise in  $\gamma$  with depth.

# Self-consistency: does the frozen background matter?

Unfreezing  $A_z^1$  on the tachyonic branch (the *most* background-coupled case), pushed to  $|a| \approx 2.5$ :

- **Linear phase validated:**  $\gamma = 1.45$  (unfrozen) vs. 1.43–1.46 (eigensolver).
- **Back-reaction confirmed:**  $\Delta A_z^1 = -0.04|a|^2$  over 4 decades; endpoint exact ( $-0.242$  vs.  $-0.242$ ).
- **Net feedback:** weakly stabilizing (amplitude ratio  $0.998 \rightarrow 0.933$ ), two orders of magnitude too weak to saturate the branch.

## Physical meaning

The frozen-battery picture is if anything an *underestimate* of the drive's robustness — the beams continuously re-supply the background rather than draining it. Saturation is by fluid cavitation, not background depletion.

## Box-size reduction: $L_x = 2\pi$ is safe, and $2.2\times$ faster

Historical box  $L_x = 6\pi$  assumed a double-tanh profile never actually used in production. 17 GPU runs (t133+abi):

- Reference anchor:  $L_x = 2\pi, 3\pi, 4\pi$  agree to 0.03% once  $\xi_{\text{sponge}}$  genuinely fits.
- Real speedup: **151s ( $6\pi$ ) vs. 69s ( $2\pi$ )** —  $2.2\times$ , not the naive  $3\times$  (fixed per-step overhead).
- Archive screen: 43% of 4192 historical points fit a  $2\pi$  box; a **tiered  $2\pi/3\pi/4\pi/6\pi$  policy** is now implemented.

### Caveat

An early “ $2\pi$  is safer” result was a sponge-miscalibration artifact (one leg had *no active sponge at all*) — resolved once redone correctly: 0.01% agreement.

# The $\alpha \geq 6$ catastrophe: partly a numerics question after all

Earlier verdict (“physical, resolution-independent”, based on amplitude and window tests only) is now qualified by a timestep test never previously run:

| Point                                         | halt (courant 0.1) | halt (courant 0.02)         | <b>Clean split:</b> |
|-----------------------------------------------|--------------------|-----------------------------|---------------------|
| $\alpha = 6, V_0 = 0.10, k_z = 5$ (xi_cut)    | $t = 31.4$         | $t = 64.5$ (2.05 $\times$ ) |                     |
| $\alpha = 6, V_0 = 0.10, k_z = 8$ (xi_cut)    | $t = 32.0$         | $t = 65.0$ (2.03 $\times$ ) |                     |
| $\alpha = 6, V_0 = 0.05, k_z = 8$ (xi_sponge) | $t = 6.0$          | $t = 6.0$ (no change)       |                     |

xi\_cut-confined catastrophes are genuinely resolution-sensitive (finer timestep  $\approx$  doubles survival, 6/7 points,  $k_z = 4$ –8); xi\_sponge-confined catastrophes at these  $\alpha$  are not (fail almost instantly regardless of resolution — a different, harder mechanism).

## Caveat

Also found:  $k_z$  within  $\sim 1$  of `kz_suppress_hi` is a real, *large* resolution artifact at  $N_Z = 64$  —  $k_z = 13$  (hi=14):  $\gamma$  nearly **doubles** at  $N_Z = 128$ . Avoid measuring there, or bump  $N_Z$ .

- The high- $\alpha$  corner's resolution uncertainty (10–12%) and the  $\alpha \geq 6$  catastrophe's numerics-dependence are both **real, open questions** — worth a dedicated study, but not on Letter A's critical path.
- **For now: tighten the  $\alpha$  range used for any headline claim** in Letter A to stay clear of the corner where resolution is uncertain and the catastrophe is only partially characterized, and revisit the wider range later if the paper's scope demands it.
- This keeps the P-collapse and confinement-set peak claims resting on the part of parameter space that is unambiguously resolved.

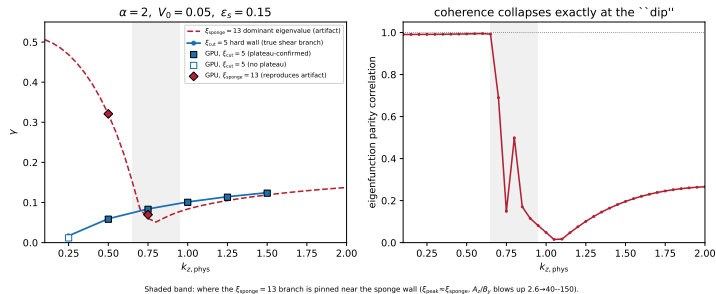
## T1.3 — closed: the claim was real as a measurement, not as physics

The original sub- $k_z = 1$  “two-branch” claim ( $\gamma = 0.32 \rightarrow 0.06$  across  $k_z = 0.5 \rightarrow 0.75$ ,  $\alpha = 2$ ,  $V_0 = 0.05$ ) came from the **pre-2026-07-17 buggy pipeline** and was retracted as unestablished. It has now been **re-derived independently of that pipeline** — pure linear algebra plus a dedicated GPU check — and **closed**: it is a sponge-boundary artifact, not a crossing between two physical eigenmode families. **Method**: a fine ( $\Delta k_{z,\text{phys}} = 0.05$ ) eigensolver continuation at  $\alpha = 2$ ,  $V_0 = 0.05$ ,  $\varepsilon_s = 0.15$ , scanning many shift-invert seeds per  $k_z$  to catch several leading branches, at three sponge radii ( $\xi_w = 8, 13, 20$ ), tracking eigenfunction peak location  $\xi_{\text{peak}}$ ,  $A_z/B_y$  ratio, and parity.

### Caveat

**Finding**: at  $\xi_w = 13$  (the original setting) the solver *does* reproduce 0.320/0.060 almost exactly — but that eigenfunction is pinned near or at the sponge radius itself ( $\xi_{\text{peak}} \approx 10\text{--}13$ ; at  $\xi_w = 8$  it locks to  $\xi_{\text{peak}} = -8.181$ , the exact wall, for every  $k_z \geq 0.85$ ), its  $A_z/B_y$  ratio blows up  $2.6 \rightarrow 40\text{--}150$ , and its parity correlation collapses  $0.99 \rightarrow <0.3$  precisely where the “dip” occurs — the signature of a sponge-supported quasi-mode, not a second confinement branch.

## T1.3 — the decisive control, and GPU confirmation



Shaded band: where the  $\xi_{\text{sponge}} = 13$  branch is pinned near the sponge wall ( $\xi_{\text{peak}} = \xi_{\text{sponge}}$ ,  $A_z/B_y$  blows up 2.6–40–150).

A hard  $\xi_{\text{cut}} = 5$  wall (well inside any sponge) gives a smooth, monotonically rising, artifact-free  $\gamma(k_z)$  — consistent with T1.2's exact-action result that the true shear branch has no interior  $k_z$  maximum, now extended below  $k_z = 1$ . **Confirmed in full nonlinear CUDA** (`run_campaign_t13confirm.sh`, `t133`): the  $\xi_{\text{cut}} = 5$  branch matches the eigensolver to 1–2% at every  $k_z \geq 0.5$  — the best GPU-vs-eigensolver agreement in the project to date. Two runs reproducing the original  $\xi_w = 13$  config reproduce the “0.32/0.06” numbers almost exactly in the *full nonlinear* simulation, not just the linear solver — the artifact is a real property of the sponge kernel, not a solver curiosity.

## Scorecard against the three blocking items

| Item | Asks                               | Status        | Evidence                                                                                                                                              |
|------|------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1.1 | Survives varying $\varepsilon_s$ ? | ✓ <b>Done</b> | $P$ -collapse, GPU-confirmed $\alpha = 0.2\text{--}6$ , $\varepsilon_s = 0.10\text{--}0.45$ , median 9.9%/21.6%                                       |
| T1.2 | Why does the peak move?            | ✓ <b>Done</b> | Exact-action reduction, ceiling, confinement-set peak; T2.5 collapse $0.977 \pm 0.011$                                                                |
| T1.3 | Sub- $k_z = 1$ branch              | ✓ <b>Done</b> | Closed as a sponge-boundary artifact (eigen-solver continuation + $\xi_{\text{cut}} = 5$ control + GPU confirmation to 1–2%) — not a genuine crossing |

**All three of Letter A's original gating items are now closed. Framing change, driven by the data:** not “the shear width is irrelevant, EPS-free” but “**coupling and confinement radius jointly select the wavelength; the flow profile does not enter directly, only through the single group**

$P = (\alpha V_0^2)^{1/3} / \varepsilon_s$ .” A sharper, still-novel claim — and the sub- $k_z = 1$  “two-branch” idea drops out of Letter A's planned content entirely, as refuted rather than confirmed.

# Where this sits in the literature

Novelty survey (LITERATURE.md, 11 targeted searches): **no prior KH / shear-driven non-Abelian plasma instability work found.**

- Closest prior art: Manuel–Mrówczyński chromohydrodynamics (colored two-fluid, but two-stream driven, no shear, no eigensolver map).
- Chromo-Weibel family (Mrówczyński et al.): anisotropy-driven, never shear-driven — our  $k_z = 0$  anchor is in this family, the shear branch is new.
- Outer branch: Nielsen–Olesen class (1978) — cite directly.
- Abelian ancestor: Das & Kaw EMHD lineage. Csernai et al.: Abelian (color-neutral) KH in heavy-ion collisions — sharply distinguish.

## Caveat

Urgency: dark-U(1) PIC (arXiv:2411.11958, 2025) flags non-Abelian dark sectors as open; a 2026 three-component BEC-KHI paper moves in the adjacent space. **INSPIRE citation-trail audit before submission** — not yet done.

## Next steps

- 1 T1.3 closed — drop the sub- $k_z = 1$  branch-crossing claim from Letter A's planned content; nothing further blocking on it.
- 2 Keep the headline  $\alpha$  range tightened to the well-resolved corner for now; revisit the high- $\alpha$ /high- $V_0$  resolution and catastrophe questions as a separate, non-blocking study.
- 3 INSPIRE citation-trail audit (Mrówczyński review  $\cap$  Csernai citers).
- 4 Draft the actual  $\sim 4$ -page Letter: model + ceiling +  $P$ -collapse + confinement-set peak + BEC-realizability paragraph + QGP/dark-sector motivation.

**All three blockers closed. The reduction is exact; its quantization matches the eigensolver to  $\leq 2\%$ , confirmed on  $> 1000$  GPU runs. Letter A drafting can proceed.**

# Thank you.

Questions, and where to push next.